

1. Administrative & Study Identification

Official Full Study Title: A phase II, multicenter, randomized, open-label study to evaluate the safety and efficacy of 400 mg of ribociclib in combination with non-steroidal aromatase inhibitors for the treatment of pre- and postmenopausal women with hormone receptor-positive, HER2-negative advanced breast cancer who received no prior therapy for advanced disease **Brief Study Title:** Study of 2 Ribociclib Doses in Combination With Aromatase Inhibitors in Women With HR+, HER2- Advanced Breast Cancer **Short Title/Acronym:** AMALEE Study **Protocol Number:** CLEE011A2207 **NCT Number:** NCT03822468 **Posting ID:** 726292173838059615 **Sponsor/ Company Name:** Novartis **Investigational Product:** Ribociclib (LEE011) + Non-Steroidal Aromatase Inhibitors (Letrozole or Anastrozole) **Phase of Development:** Phase II **Trial Status:** COMPLETED (Study has been completed) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate whether a 400 mg ribociclib starting dose can demonstrate noninferiority in Overall Response Rate (ORR) compared to the standard 600 mg starting dose, while reducing the incidence of adverse events (AEs). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Clinical Benefit Rate (CBR), Duration of Response (DOR), Time to Response (TTR), Overall Survival (OS), pharmacokinetics (Cmax, AUC), and the safety/tolerability profile (e.g., QTcF prolongation, neutropenia).

2.2 Background & Rationale To address the gap in tolerability and treatment adherence for patients receiving CDK4/6 inhibitors. The standard 600 mg ribociclib starting dose is highly effective but associated with dose-dependent adverse events (such as neutropenia and QTcF prolongation) that frequently require dose modifications. This study evaluates if a lower starting dose (400 mg) maintains clinical efficacy while improving the patient safety profile.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (600 mg Ribociclib standard dose) **Blinding:** Open-label **Randomization:** 1:1 Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 376
evaluable patients **Number of Sites:** 107 sites across 23 countries
(Global) **Estimated Study Start:** 18-Jun-2019 **Estimated Primary
Completion:** 30-Aug-2024 **Current Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Pre- and postmenopausal women physiologically capable of undergoing treatment. 2. Histologically and/or cytologically confirmed diagnosis of ER-positive and/or PgR-positive, HER2-negative advanced breast cancer (loco-regionally recurrent or metastatic). 3. Measurable disease, i.e., at least one measurable lesion according to RECIST version 1.1. 4. Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1.

4.2 Exclusion Criteria 1. Received any prior systemic anti-cancer therapy for advanced breast cancer (prior neo-/adjuvant therapy allowed). 2. Symptomatic visceral disease or any disease burden that makes the patient ineligible for endocrine therapy. 3. Clinically significant, uncontrolled heart disease and/or cardiac repolarization abnormality (e.g., QTcF interval $\geq$ 450 ms at screening).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** Ribociclib 400 mg QD on Days 1 to 21 of a 28-day cycle, followed by 7 days off. * **Active Comparator:** Ribociclib 600 mg QD on Days 1 to 21 of a 28-day cycle, followed by 7 days off. * **Combined with:** Letrozole 2.5 mg QD continuously OR anastrozole 1 mg QD continuously. Premenopausal women also receive **goserelin** (Trade Name: **Zoladex**; ATC Code: **L02AE03**) 3.6 mg Q4W. **Route of Administration:** Oral (ribociclib, letrozole, anastrozole); Subcutaneous injection (goserelin). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or discontinuation from the study treatment for any other reason.

5.2 Concomitant & Prohibited Therapy Permitted: LHRH agonists (goserelin) for premenopausal women. **Prohibited:** Concurrent anti-cancer therapy; systemic corticosteroids $\leq$ 2 weeks prior to starting study drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Triplicate ECG, Serum Pregnancy Test, RECIST baseline
Baseline	Day 0	Randomization, First Dose	
Interim	Every 28-Day Cycle	Safety Labs, Efficacy Assessment,	

ECGs (e.g., Cycle 1 Day 15 for QTcF monitoring), Pharmacokinetics
| | **End of Study** | Variable | Treatment Discontinuation, Final
Vital Signs, Post-treatment Efficacy Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Response Rate (ORR) – to demonstrate noninferiority of the 400 mg ribociclib dose compared to the 600 mg dose. **Measurement Method:** Investigator assessment using Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS), Clinical Benefit Rate (CBR), and Duration of Response (DOR). * Time to Response (TTR) and Overall Survival (OS). * Change from baseline in QTc (with Fridericia's correction) at Cycle 1 Day 15 (at 2 hours post-dose). * Pharmacokinetics (Maximum plasma concentration after dose and 24-hour area under the curve) and safety metrics (rates of Grade 3/4 neutropenia and QTcF prolongation).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine laboratory testing for neutropenia, liver function, and centralized ECG monitoring for QTcF prolongation (assessing Δ QTcF at Cycle 1 Day 15). **Serious Adverse Events (SAEs):** Reported standardly; specific focus on interstitial lung disease, pneumonitis, and kidney toxic effects. **Data Monitoring Committee (DMC):** Internal and external safety monitoring conforming to Novartis standards and GCP.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome (ORR); Safety Set for the incidence of AEs. **Sample Size Justification:** $N=376$ powered to test the noninferiority margin of the experimental 400 mg dose against the 600 mg active control. **Handling of Missing Data:** Standard imputation strategies for missing tumor assessments and pharmacokinetic data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical monitoring, centralized 12-lead ECG review, and pharmacokinetics tracking. **Audit Trail:** eCRF tracking and data validation of local laboratory results versus centralized assessments.

11. Study Identifiers & Governance

Primary ID: CLEE011A2207 **Registry Number:** NCT03822468 **Posting ID:** 726292173838059615 **Sponsor/Company:** Novartis **Brief Study Title:** Study of 2 Ribociclib Doses in Combination With Aromatase Inhibitors in Women With HR+, HER2- Advanced Breast Cancer **Official Regulatory Title:** A phase II, multicenter, randomized, open-label study to evaluate the safety and efficacy of 400 mg of ribociclib in combination with non-steroidal aromatase inhibitors for the treatment of pre- and postmenopausal women with hormone receptor-positive, HER2-negative advanced breast cancer who received no prior therapy for advanced disease

12. Clinical Framework (Breast Cancer Specific)

Disease Area: Breast Cancer **Preferred UMLS Name:** Hormone Receptor Positive HER2 Negative Breast Cancer **Diagnosis Threshold:** Histologically/cytologically confirmed ER+/PgR+ and HER2- **Medical Methodology:** CDK4/6 inhibitor in combination with Endocrine Therapy **Optimization Target:** Ribociclib starting dose optimization (400 mg vs 600 mg) for AE management

13. Study Procedures & Methodology

Management Pathway: First-line systemic therapy for HR+/HER2- advanced breast cancer **Education Protocol (HCP):** Investigator training on dose modifications and proactive monitoring for neutropenia/QT prolongation **Education Protocol (Patient):** Adherence counseling on the 21-days-on / 7-days-off oral dosing schedule and AE self-reporting **Quality Audit Mechanism:** Centralized evaluation of triplicate ECGs and pharmacokinetic exposure parameters

14. Observation & Follow-Up Schedule

Total Duration: ~5 years (Trial initiated Jun 2019, final data cutoff Aug 2024) **Onsite Visit Cadence:** Evaluated in 28-day treatment cycles **Remote Monitoring Frequency:** Continuous monitoring of safety signals per cycle **Checklist Review Interval:** Regular tumor progression assessments per RECIST 1.1 guidelines

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				